

Detailed Session Breakdown

- ✂ **Instructor:** Yousef Khaled Omar
 - ✂ **Target Audience:** Beginners in Embedded Systems
 - ✂ **Tools & Hardware:** STM32CubeIDE, STM32 development board, sensors, actuators
 - ✂ **Duration:** 3-4 hours
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◇ Session 1: Introduction to Embedded Systems

✂ **Goal:** Provide an overview of embedded systems, introduce ARM architecture, and set up STM32CubeIDE.

Topics Covered:

☒ What is Embedded Systems?

- Embedded system vs computing system
- Difference between microcontrollers and microprocessors
- Examples of embedded systems in real life (smartphones, cars, IoT)

☒ Understanding the General Microprocessor Architecture

- Introduction to microcontrollers Architecture (register file, Control Unit, ALU ...)
- Fetch-Decode-Execute Cycle (How a microcontroller runs code)
- Instruction set (RISC Vs CISC)
- Processor Architecture (Von neuman vs Harvard)
- Memory organization in microcontrollers (RAM Vs ROM)
- memory layout (.bss .rodata stack...)
- RAM (SRAM Vs DRAM)
- ROM (Masked ROM, OTPROM, EPROM, EEROM and Flash)
- Tool chain (preprocess, compiler, assembler and linker)
- Flashing (Off circuit vs On circuit)
- Embedded Software Architecture (APP, HAL LIB AND LL)

☒ Understanding the ARM Architecture

- Introduction to ARM (ARM history, what ARM stands for)
- ARM Architecture (voltage regulator, PIO and communication peripherals)
- Intro to AMBA (AHB vs APB, Peripheral bridge)
- Power management (RCC vs Enable)

Introduction to STM32CubeIDE

- Why STM32 is widely used in the industry
- Installing and setting up STM32CubeIDE
- Creating a new STM32 project
- Introduction to HAL (Hardware Abstraction Layer) and LL (Low Level) drivers

Practical:

- Running an built-in example in STM32cube
- We will assign them a task that let theem write a summary about the lecture in 1 page pdf.

Conclusion:

- We decided to assign a small task which is just to make a 1 page PDF in which you just look at each slide and give one information you learned about it ex:
RAM is fast, ROM is slow
This task will make the students re-open the slides to study it and will learn them documentation skill.
- We preferred this session to be offline.
- We made a pre-session requirements like
Installing STM32 and protous
Sending them a short video that covers our topics
You will find the links in the last slide